

Installing and testing 15LSS_sample_gas on the IOC host

The IOC serves `15IDC:SampleGas:` and drives the Alicat `15IDC:Alicat1:` from the O2 reading `15IDC:D1Dmm_calc`. It installs into the ChemMat area of the synApps tree, `/home/chem_epics/chemmatCARS/synApps/support/ChemMat`.

Read this first. A normal start (`start_ioc`) is *live*: if helium is already flowing (O2 below 10 % and the Alicat setpoint above 0) it takes over the Alicat within seconds. So the first start below is a shadow start, which writes nothing. Stop the old Phoebus timer script before you switch to live: two writers fight over the setpoint.

1. Get the code

Clone only the IOC and its screens from GitHub:

```
cd /home/chem_epics/chemmatCARS/synApps/support/ChemMat
git clone --sparse --filter=blob:none \
  https://github.com/Dave-J-W/helium-flow-fb.git lssSampleGas
cd lssSampleGas
git sparse-checkout set ioc/lssSampleGas ioc/screens
git log -1 --online          # the version you install
```

The IOC top is then `/home/chem_epics/chemmatCARS/synApps/support/ChemMat/lssSampleGas/ioc/lssSampleGas`, and the screens are in `/home/chem_epics/chemmatCARS/synApps/support/ChemMat/lssSampleGas/ioc/screens`. If git says `unknown option --sparse`, it is older than 2.27; leave out `--sparse --filter=blob:none` and the `sparse-checkout` line. That clones the whole repo (about 15 MB), and everything else stays the same.

2. Build

```
cd /home/chem_epics/chemmatCARS/synApps/support/ChemMat/lssSampleGas/ioc/lssSampleGas
make 2>&1 | tee build.log
lssSampleGasApp/src/0.linux-x86_64/sgUnitTest | tail -1      # PASS: 45/45
lssSampleGasApp/src/0.linux-x86_64/sgIocTest  | tail -1     # PASS: 30/30
```

No configuration is needed: `configure/RELEASE` already has this host's paths (base `/usr/local/epics/base`, seq R2-2-9, asyn R4-44-2, calc R3-7-5, sscan R2-11-6, std R3-6-4, autosave R5-11). The same versions build without warnings on gcc 11 (checked on Ubuntu 22.04). If `make` fails, send the last 40 lines of `build.log`.

3. First start: shadow, in a terminal

Check that no `15LSS_sample_gas` is already running under procServ, then:

```
cd /home/chem_epics/chemmatCARS/synApps/support/ChemMat/lssSampleGas/ioc/lssSampleGas
cd iocBoot/iocLSS_sample_gas
./startLSSSampleGasTest
```

This is the production IOC in **shadow mode**: it reads everything and writes nothing. The console is this terminal; type `exit` to stop it. Its settings go to `autosave-test/`, so nothing from the test carries over to the production start. Within a few seconds you should see:

- PV names: MFC 15IDC:Alicat1:, O2 15IDC:D1Dmm_calc, CYL none

- `start in shadow mode (FORCE_SHADOW): Par:writeEnable set to 0`
- a state line, e.g. `→ IDLE (restart: setpoint is 0 ($4.7))`

If you see `waiting for the Alicat PVs ...` instead, the Alicat IOC isn't up or its PV names differ; the controller keeps waiting and decides once they connect.

From another terminal, compare with the O2 and Alicat panels you already have:

```
caget 15IDC:SampleGas:Sts:O2 15IDC:SampleGas:Sts:Flow 15IDC:SampleGas:Sts:SetpointRBV
caget 15IDC:SampleGas:Sts:State 15IDC:SampleGas:Sts:TickAge
```

`Sts:TickAge` should be 0 or 1 (seconds since the controller last ticked).

4. Screens

The displays are in `/home/chem_epics/chemmatCARS/synApps/support/ChemMat/lssSampleGas/ioc/screens`. Open `sampleGas_simple.bob` (everyday) or `sampleGas_main.bob` (full panel) with the macro `P=15IDC:SampleGas:`, e.g. from an action button in one of your menu displays ("Open display", macros `P = 15IDC:SampleGas:`). The displays open each other and pass `P` on. `docs/ioc/OPERATOR_GUIDE.md` in the repo explains every field.

In shadow mode both panels show a SHADOW MODE badge and the banner shows the MAJOR shadow mode: the controller is not writing to the Alicat.

5. Test sequence

Have helium on and the enclosure closed. **Flow Zero** stops the helium at any point.

1. **Watch in shadow** for a few minutes: readings match, `Sts:TickAge` stays 0–1.
2. **Stop the old timer script.**
3. **Go live:** Admin → **Live (on)** → confirm. It goes to IDLE; the valve doesn't move.
4. **Take over: Purge** (from air: purge at 20 SLPM, lid check, handoff, regulation in about 6–7 minutes), or **Resume Flow** if the box is already purged (O2 below 10 %).
5. **Regulate:** O2 settles at the target (0.99 %), in-range LED green.
6. **Flow Zero:** the Alicat goes to 0.
7. **Finish:** `exit` in the IOC terminal. The Alicat keeps its last setpoint.

6. Production start

1. Add the `start_ioc` line (check port 20125 is free in `IOCLIST`). It is one line; in the PDF it only wraps because of the page width:

```
15LSS_sample_gas 20125 1
/home/chem_epics/chemmatCARS/synApps/support/ChemMat/lssSampleGas/ioc/lssSampleGas/iocBoot/iocLS
S_sample_gas/startLSSsampleGas
```

2. Keep the old timer script off, and make sure the test IOC has exited.
3. `start_ioc 15LSS_sample_gas`. It is live. With the Alicat at 0 it waits in IDLE for Purge or Resume Flow; with helium flowing and the box purged it resumes regulating at once.
4. Alarm server: load `ioc/screens/sampleGas_alarms.xml`. The archiver PV list is `ioc/screens/archive_pvs.txt`.

After a crash (procServ does not restart it), start it again with `start_ioc`; the Alicat holds its setpoint meanwhile and the restart resumes without a bump.

7. Updating

```
cd /home/chem_epics/chemmatCARS/synApps/support/ChemMat/lssSampleGas
git pull
cd ioc/lssSampleGas && make 2>&1 | tee build.log
```

Then restart the IOC (`start_ioc 15LSS_sample_gas`, or `exit` in its console). `git pull` keeps the IOC's settings (autosave/) and logs; they are not in the repo.

Troubleshooting

You see	What to do
waiting for the Alicat PVs (...): controller not acting yet	The Alicat IOC is down, or its PV names differ: <code>caget 15IDC:Alicat1:Setpoint_RBV</code> . The names can be changed on Deep admin.
not configured: <code>Cfg:... is empty</code>	A PV name on Deep admin is empty.
MFC gas table is ..., not He	Set the Alicat's gas table to helium.
MFC flow units are ..., not SLPM	Set the Alicat's flow units to SLPM.
flow mismatch: cylinder empty or MFC fault?	The flow doesn't follow the setpoint: cylinder valve, pressure, regulator.
Alicat setpoint ... does not follow the controller	A write was lost or something else writes the setpoint (the old timer script?).
controller not ticking (red over the banner)	The IOC answers but the controller stopped: restart the IOC.
Screens all magenta	The IOC isn't running, or Phoebus can't reach it (CA address list).